

Out-of-equilibrium ceRNA crosstalk

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Abstract Amongst non-coding RNAs, microRNAs are pivotal post-transcriptional regulators of gene expression in higher eukaryotes. Through a titration-based mechanism of interaction with their target RNAs, microRNAs can mediate a weak but pervasive form of RNA cross-regulation, as different endogenous RNAs can be effectively coupled by competing for microRNA binding (a phenomenon now known as ‘crosstalk’). Mathematical modelling has been proven of great help in unraveling many features of these competing endogenous RNA (ceRNA) interactions. However, although many studies have been devoted to the steady-state properties of this indirect regulatory layer, little is known about how the information encoded in frequency, amplitude, duration, and other features of regulatory signals can affect the resulting ceRNA crosstalk picture and hence the overall patterns of gene expression. Here we focus on such dynamical aspects, with a special emphasis on the encoding and decoding of time-dependent signals.

Key words: miRNA, ceRNA, crosstalk, out-of-equilibrium, frequency preference

1 Introduction

About fifteen years ago, a burst of research works revolutionised the way we think about gene regulation. Their common denominator was the fact that several non-directly interacting RNA gene products were indeed responding to each others' variation, in terms of overexpression or downregulation, in a non-coding dependent way [15, 46, 66, 72, 74]. The origin of such couplings was found in a peculiar mechanism of competition: these RNAs shared some microRNA (miRNA) regulators, whose binding they were actually competing for. Since then, the term Competing Endogenous RNA (ceRNA) has been used to identify all those RNA molecules (coding and non-coding) that endogenously compete for a shared pool of miRNAs and can in turn crossregulate each other [41, 53, 57, 67, 77].

MiRNAs are short non-coding RNAs (ncRNAs) that regulate eukaryotic gene expression post-transcriptionally [7, 14, 34, 42]. Their biogenesis can be summarized as follows. Once in their mature stage within the cytoplasm, miRNAs get uploaded into multiprotein RNA-induced silencing complexes (RISC) [38] and then recognize their RNA targets via sequence specific binding [6]. The miRNA-target interaction takes place through Watson-Crick base pairing of the miRNA to ~ 8 -nucleotide long words on the target RNAs (the so-called miRNA response elements, or MREs), and can result in either an increase in the target turnover or in the repression of its translation. Which path is chosen depends on the type of target (coding or non coding) and on the degree of complementarity of the two sequences [17, 18, 24, 45], but the net effect most of the times is a downregulation of the target expression. Together with mRNAs, miRNA targets include different sorts of ncRNAs, like lncRNAs, circRNAs and pseudogenes [25, 40, 43]. The fact that miRNA-based regulation is combinatorial, i.e. each miRNA can regulate the expression of multiple RNA targets and each target can be simultaneously regulated by several miRNAs [50, 69, 70], results in a great complexity of the miRNA-RNA interaction network [61]. miRNAs play fundamental roles in different pathologies, ranging from cancer to neurodegenerative diseases, and in key processes concerning cell differentiation and development [2, 5, 12, 28, 44, 55, 56, 78]. The tissue-specific expression of miRNAs and their conservation across higher eukaryotes, as well as the conservation of their mRNA target structures, suggest they are pivotal regulators of protein repertoire composition and cell specificity [5, 8, 26, 35, 54].

One of the possible role of miRNAs is related to the fine-tuning of the expression levels of their targets, i.e. to the efficient control of fluctuations in gene expression [68]. This is supported by the over-representation, in the miRNA-RNA interaction network, of several miRNA-mediated motifs [19], including incoherent feedforward loops and toggle switches [63, 64], whose theoretical [9, 39, 63] and experimental [47, 71] analysis showed target noise dampening. However, while this seems to be true when miRNAs are embedded in peculiar circuitries, recent studies showed that miRNA may also amplify their targets' variability, with cells exploiting this to easily explore different developmental gene expression programs [16, 21, 36].

More than a decade ago, however, it has been shown that the miRNA-target interaction is compatible with a titration-like mechanism, which can in turn generate

thresholds in target gene expression [62]. Threshold-like behaviours are a signature of hypersensitivity regions [13, 27] where the correlation between targets is maximal and their relative stoichiometry highly coupled via retroactivity [22, 65], i.e. via the correlation between their intrinsic and extrinsic fluctuations [73]. In a system with many miRNAs sharing multiple RNA targets, such a hypersensitivity generates specific ranges of relative miRNA-RNA stoichiometries, identified by a “quasi-equimolarity condition” [1], that allow for maximal crosstalk (i.e. positive cross-correlation) between miRNA targets [11]. In other words, the correlation patterns among different RNA species are regulated to some degree by the relative abundances of miRNAs and their targets [10, 31].

Such relative abundances allow for the definition of three different stoichiometric regimes of miRNA-target interactions: (i) a “repressed regime”, where miRNA levels greatly exceed target levels (so that most of the targets are repressed); (ii) a “susceptible regime”, where the two levels are comparable (and fluctuations of the different molecular species are highly coupled); and (iii) an “unrepressed regime”, where the abundance of targets greatly overcomes that of miRNAs (so that most of the targets are expressed) [58]. Moreover, the strength of ceRNA crosstalk is different if measured on different time-scales, e.g. longer time scales (or equilibrium) versus shorter time scales (transient) [30]. Figure 1 shows a simple *in silico* example in which two different RNA species can crosstalk through a shared miRNA. Here, the amount of ceRNA 1 (m_1 , in red) is changed at a given time point within the transient (left panels) or at the steady state (right panels) in the three different stoichiometric regimes. As is clear from the figure, the stationary level of ceRNA 2 (m_2 , in blue) changes in response to a perturbation of the transcription rate of ceRNA 1 when the latter is in the susceptible or repressed regimes. On the other hand, transient changes in ceRNA levels are observed also when both ceRNAs are unrepressed at stationarity, suggesting that some form of crosstalk is active during the approach to equilibrium independently of whether steady state levels are responsive or not.

Figure 1 also suggests that the way ceRNAs respond to changes in each other’s transcription levels may also depend on the characteristics of the generic input signal, i.e. on whether it is constant (as in Figure 1) or pulsed, and on which node of the regulatory circuit this signal is applied to (i.e. on whether it affects the expression of the miRNA or of one of the ceRNAs). Reviewing the quantitative grasp of this complex phenomenology obtained through mathematical models is the goal of this chapter. We will start with a brief recap of the minimal model of miRNA-ceRNA network on which we shall focus, and of the type of temporal dynamics that a system of ceRNAs can develop as the various time scales involved change.

2 Overview of a minimal ceRNA model and its relevant time scales

We focus on the minimal model of the miRNA-ceRNA network depicted in Figure 2A.

For each node of a bipartite system of M miRNAs interacting with N ceRNAs we take into account the essential features of transcription and degradation; each link represents the interaction between a miRNA (here intended as a mature miRNA already uploaded into the RISC) and a ceRNA, and takes into account the events of binding and unbinding of the miRNA-ceRNA complex and its processing. Each of these reaction events is represented in more detail in Figure 2B, with the associated kinetic rates. ceRNA (resp. miRNA) species are indexed by i (resp. a) and their levels are denoted by m_i (resp. μ_a). The level of miRNA-ceRNA complexes is instead denoted by c_{ia} . The relevant rates are given by the transcription rates of miRNAs (β_a) and ceRNAs (b_i), their degradation rates (δ_a and d_i respectively), and the miRNA-ceRNA association (k_{ia}^+) and dissociation (k_{ia}^-) rates. In addition, miRNA-ceRNA complexes can undergo different fates [3, 75], namely stoichiometric degradation (rate σ_{ia}), whereby the entire complex is degraded, or catalytic degradation (rate κ_{ia}), where the miRNA is recycled.

Adopting the notation used in [58], the mass-action kinetic equations describing the above processes read

$$\begin{aligned}\frac{dm_i}{dt} &= b_i - m_i d_i - \sum_a k_{ia}^+ m_i \mu_a + \sum_a k_{ia}^- c_{ia} , \\ \frac{d\mu_a}{dt} &= \beta_a - \mu_a \delta_a - \sum_i k_{ia}^+ m_i \mu_a + \sum_i (k_{ia}^- + \kappa_{ia}) c_{ia} , \\ \frac{dc_{ia}}{dt} &= k_{ia}^+ m_i \mu_a - (\sigma_{ia} + \kappa_{ia} + k_{ia}^-) c_{ia} ,\end{aligned}\tag{1}$$

where the indices i and a range from 1 to N and from 1 to M , respectively. The steady state, which is unique and asymptotically stable [33], is given by the solutions of

$$\begin{aligned}\langle m_i \rangle &= \frac{b_i + \sum_a k_{ia}^- \langle c_{ia} \rangle}{d_i + \sum_a k_{ia}^+ \langle \mu_a \rangle} , \\ \langle \mu_a \rangle &= \frac{\beta_a + \sum_i (k_{ia}^- + \kappa_{ia}) \langle c_{ia} \rangle}{\delta_a + \sum_i k_{ia}^+ \langle m_i \rangle} , \\ \langle c_{ia} \rangle &= \frac{k_{ia}^+ \langle \mu_a \rangle \langle m_i \rangle}{\sigma_{ia} + \kappa_{ia} + k_{ia}^-} ,\end{aligned}\tag{2}$$

where $\langle x \rangle$ represents the steady state level of the variable x . If one eliminates $\langle c_{ia} \rangle$ from (2), the much simpler conditions

$$\langle m_i \rangle = \frac{m_i^*}{1 + \sum_a \langle \mu_a \rangle / \mu_{0,ia}} ,\tag{3}$$

$$\langle \mu_a \rangle = \frac{\mu_a^*}{1 + \sum_i \langle m_i \rangle / m_{0,ia}} .\tag{4}$$

are obtained, where $m_i^* \equiv b_i/d_i$ and $\mu_a^* \equiv \beta_a/\delta_a$ are the constitutive maximal values achievable by the ceRNA and miRNA species in absence of interactions, whereas

$$m_{0,ia} = \frac{\delta_a}{k_{ia}^+} \left(1 + \frac{k_{ia}^- + \kappa_{ia}}{\sigma_{ia}} \right), \quad (5)$$

$$\mu_{0,ia} = \frac{d_i}{k_{ia}^+} \left(1 + \frac{k_{ia}^-}{\sigma_{ia} + \kappa_{ia}} \right), \quad (6)$$

represent reference concentration values which depend on the specific miRNA-ceRNA pair. These values are the “thresholds” depicted in Figure 2C. Equations (3) and (4) have been analyzed in depth in different regimes and for different types of networks, from small modules to genome-scale interactomes [31, 60, 61]. Their usefulness lies in the fact that they succinctly summarize the three stoichiometric regimes briefly described in the introduction:

- **unrepressed (or free) regime:** when $\sum_a \mu_a / \mu_{0,ia} \ll 1$, $\langle m_i \rangle$ is close to its constitutive value m_i^* , so that changes in miRNA transcription rates will not affect the asymptotic abundance of ceRNA i ;
- **susceptible regime:** when $\sum_a \mu_a / \mu_{0,ia} \simeq 1$, $\langle m_i \rangle$ decreases sigmoidally as the quantity $\sum_a \mu_a / \mu_{0,ia}$ increases (for example due to the increase in the level of one miRNA species). In this condition, the steady state level of ceRNA i is highly susceptible to changes in miRNA levels;
- **repressed regime:** when $\sum_a \mu_a / \mu_{0,ia} \gg 1$, $\langle m_i \rangle$ is very small: ceRNA i is fully repressed.

It is clear that these regimes are well defined at the steady state and are regulated by soft threshold values that depend (i) on the set of parameters that define the constitutive miRNA and ceRNA values, and (ii) on the kinetic parameters that appear in (1).

To quantify the magnitude of ceRNA crosstalk at stationarity, one can employ the *susceptibility*

$$\chi_{ij} = \frac{\partial \langle m_i \rangle}{\partial b_j}, \quad (7)$$

which measures the change in the steady-state level of ceRNA i that occurs in response to a perturbation in the transcription rate of ceRNA j . Based on the above discussion, one expects χ_{ij} to be non-negligible when both ceRNAs i and j are susceptible. This is indeed the case. In addition, however, χ_{ij} turns out to be non-zero also when ceRNA i is susceptible but ceRNA j is repressed. The reason is that, due to their kinetic parameters, repressed ceRNAs are excellent miRNA sponges: as soon as their transcription rate is increased, they attract more miRNAs, thereby lifting repression from ceRNA i (provided it is susceptible to changes in miRNA levels). The steady-state crosstalk scenario arising from these considerations has been reviewed in detail in [58] and has been further explored in [61].

Kinetic parameters however also encode the relevant time scales of the dynamics and define how fast the three different steady state regimes are reached. (For instance, the dynamics of the quantity c_{ia} equilibrates with a characteristic time of the order of $(\sigma_{ia} + \kappa_{ia} + k_{ia}^-)^{-1}$.) Figure 3 shows a sample of possible time trajectories in the simple case of a network with one miRNA and two ceRNAs. These

time trajectories are grouped according to their steady-state regimes (columns) and differ by the average lifetimes of the RNA species (rows A to C) and by the average catalytic degradation time for the two complexes (rows D and E). One sees that, along with the steady state levels, the dynamics also depends strongly on parameters (see e.g. the fact that some trajectories approach the steady state regularly from below, while others over-shoot the steady state during a transient and then equilibrate to it from above). Perturbations such as changes in transcription rates are therefore bound to affect the system differently, depending on when (during the dynamics) and how they occur. More generally, while the steady-state assumption may be justified in laboratory conditions, it is far from obvious that systems of ceRNAs occurring in physiological situations are stationary. As a consequence, insights on ceRNA crosstalk derived within steady state models might be of limited practical relevance.

3 Transient ceRNA response to perturbations: quantitative theory

3.1 Steady-state vs transient crosstalk

As said above, ceRNA response patterns are going to strongly depend on the type of perturbations one considers (e.g., on their frequency, magnitude, on when they are performed, etc.) as well as on the characteristic timescales of the dynamics and on how they are hierarchically organised. Such timescales can in part be read-off from (1). They include

- the mean lifetime of miRNAs ($1/\delta_a$) and ceRNAs ($1/d_i$) in absence of miRNA-ceRNA interactions;
- the timescales over which miRNA-ceRNA complexes are processed, namely the bare mean lifetime of a complex ($1/(\kappa_{ia} + \sigma_{ia} + k_{ia}^-)$) and the mean time required for a complex to be degraded either catalytically or stoichiometrically ($1/(\kappa_{ia} + \sigma_{ia})$).

In addition to these, two other timescales play a very important role to characterize the response to perturbations, namely the mean times required for a complex to be degraded catalytically ($1/\kappa_{ia}$) or, respectively, stoichiometrically ($1/\sigma_{ia}$). Furthermore, dynamical crosstalk patterns are bound to be very different from those found at stationarity. This is most evident by focusing on the role of the stoichiometric degradation of miRNA-ceRNA complexes. (To avoid the notational complications that arise in systems with multiple miRNA species, we shall henceforth consider a system with N ceRNA species but one miRNA species only.) A key aspect of the steady-state crosstalk scenario briefly outlined above is that it relies entirely on the assumption that $\sigma_i > 0$ [31] (a particularly useful theoretical description being possible when σ_i is sufficiently large, specifically $\sigma_i \gg k_i^- + \kappa_i$ [60]). This is because the equation for miRNA levels at steady state (see (2)) can be re-cast as

$$\langle \mu \rangle \left(\delta + \sum_i \frac{\sigma_i b_i}{\sigma_i + \kappa_i} \frac{1}{\langle \mu \rangle + \mu_{0,i}} \right) = \beta . \quad (8)$$

One sees that if $\sigma_i = 0$ for all ceRNAs, then $\langle \mu \rangle = \beta / \delta$ and $\frac{\partial \langle \mu \rangle}{\partial b_j} = 0$. But this implies that the susceptibility χ_{ij} (see (7)) is also zero, since

$$\chi_{ij} \equiv \frac{\partial \langle m_i \rangle}{\partial b_j} = \frac{\partial \langle m_i \rangle}{\partial \langle \mu \rangle} \frac{\partial \langle \mu \rangle}{\partial b_j} = 0 . \quad (9)$$

Therefore, no crosstalk can occur at steady state unless miRNAs are at least partially recycled when the miRNA-ceRNA complex is degraded. Obviously this needs no longer be true away from stationarity, namely when (8) does not hold. And we shall see that indeed crosstalk can occur over short timescales even when $\sigma_i = 0$ for all i .

3.2 Dynamical characterization of the ceRNA effect: small perturbations

The simplest dynamical model that can be considered is that in which the transcription rates appearing in the kinetic equations (1) have an exogenous, time-dependent component that mimics, for instance, slow variations resulting from regulatory adjustments, sudden changes due e.g. to transfections, or even periodic transcriptional pulses (that can be realized in the laboratory for instance via optical means). In such cases,

$$b_i \equiv b_i(t) = b_{i,0} + h_i(t) , \quad (10)$$

$$\beta \equiv \beta(t) = \beta_0 + h_\mu(t) , \quad (11)$$

where $b_{i,0}$ and β_0 are constants while $h_i(t)$ and $h_\mu(t)$ represent the time-dependent perturbations. Again in the simplest case, these perturbations are assumed to be “small” (i.e. such that they only cause molecular levels to deviate from the steady state by an amount that is linear in the magnitude of the perturbation). The assumption that only small deviations from the steady state occur can be exploited to derive equations for the time evolution of quantities

$$\begin{aligned} \Delta_i &\equiv m_i - \langle m_i \rangle , \\ \Delta_\mu &\equiv \mu - \langle \mu \rangle , \\ \Delta_{i\mu} &\equiv c_i - \langle c_i \rangle , \end{aligned} \quad (12)$$

that measure the divergence of molecular levels from their stationary values. (Here, c_i stands for the level of the complex formed by the miRNA and by ceRNA i .) Indeed, close to the steady state the mass-action dynamics (1), which we write compactly in vector form as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \quad (\mathbf{x} = (\{m_i\}, \mu, \{c_i\})) , \quad (13)$$

is well described by its linearized version

$$\dot{\mathbf{x}} \simeq \mathbf{f}(\langle \mathbf{x} \rangle) + \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\mathbf{x}=\langle \mathbf{x} \rangle} \cdot (\mathbf{x} - \langle \mathbf{x} \rangle) . \quad (14)$$

Upon computing the Jacobian of $\mathbf{f}$, one finds that the quantities (12) evolve according to

$$\frac{d\Delta_i}{dt} = h_i - d_i\Delta_i - k_i^+(\Delta_i\langle\mu\rangle + \Delta_\mu\langle m_i\rangle) + k_i^-\Delta_{i\mu} , \quad (15)$$

$$\frac{d\Delta_\mu}{dt} = h_\mu - \delta\Delta_\mu - \sum_i k_i^+(\Delta_i\langle\mu\rangle + \Delta_\mu\langle m_i\rangle) + \sum_i (k_i^- + \kappa_i)\Delta_{i\mu} , \quad (16)$$

$$\frac{d\Delta_{i\mu}}{dt} = k_i^+(\Delta_i\langle\mu\rangle + \Delta_\mu\langle m_i\rangle) - (k_i^- + \sigma_i + \kappa_i)\Delta_{i\mu} . \quad (17)$$

(We have assumed for simplicity that $h_\mu = 0$, see (11), so as to consider only time-dependent perturbations to ceRNA transcription rates.) Systems of this type are conveniently analyzed in Fourier space [23]. Denoting by $\hat{X}$ the Fourier transform of X , i.e. $\hat{X}(\omega) = \int X(t) \exp(-i\omega t) dt$, and applying elementary Fourier transform rules [76], one can re-cast the above equations as

$$\begin{aligned} i\omega\hat{\Delta}_i &= \hat{h}_i - d_i\hat{\Delta}_i - k_i^+(\hat{\Delta}_i\langle\mu\rangle + \hat{\Delta}_\mu\langle m_i\rangle) + k_i^-\hat{\Delta}_{i\mu} , \\ i\omega\hat{\Delta}_\mu &= \hat{h}_\mu - \delta\hat{\Delta}_\mu - \sum_i k_i^+(\hat{\Delta}_i\langle\mu\rangle + \hat{\Delta}_\mu\langle m_i\rangle) + \sum_i (k_i^- + \kappa_i)\hat{\Delta}_{i\mu} , \\ i\omega\hat{\Delta}_{i\mu} &= k_i^+(\hat{\Delta}_i\langle\mu\rangle + \hat{\Delta}_\mu\langle m_i\rangle) - (k_i^- + \sigma_i + \kappa_i)\hat{\Delta}_{i\mu} . \end{aligned} \quad (18)$$

The quantity of interest to characterize the transient ceRNA response to perturbations in ceRNA transcription rates is the dynamical counterpart of (7), i.e.

$$\hat{\chi}_{ij}(\omega) = \frac{\partial \hat{\Delta}_i(\omega)}{\partial \hat{h}_j(\omega)} . \quad (19)$$

In this setting, $\hat{\chi}_{ij}(\omega)$ measures the magnitude of the ceRNA effect over timescales of the order of $1/\omega$ (so that, for instance, $\hat{\chi}_{ij}(\omega) \simeq 0$ for $\omega \rightarrow \infty$ implies that no ceRNA effect takes place over very short timescales). The steady-state scenario is retrieved in the limit $\omega \rightarrow 0$ (implying for example that, in general, $\hat{\chi}_{ij}(\omega)$ can remain finite as $\omega \rightarrow 0$ only if $\sigma_i > 0$). Equations (18) are formally identical to the steady state conditions (2), except for the fact that they hold in Fourier space. Hence, just as one can compute the static susceptibility (9) from (2), the above equations allow to calculate quantities like (19). Such an approach has been developed thoroughly in [30]. Here we limit ourselves to outlining the key results obtained therein.

Because

$$\hat{\chi}_{ij}(\omega) = \frac{\partial \hat{\Delta}_i(\omega)}{\partial \hat{\Delta}_\mu(\omega)} \frac{\partial \hat{\Delta}_\mu(\omega)}{\partial \hat{h}_j(\omega)} \equiv g_{i\mu}(\omega) g_{\mu j}(\omega) , \quad (20)$$

the transient response is the product of two ‘gains’, one describing how the level of ceRNA i changes in response to changes in the miRNA level ($g_{i\mu}$), the other describing how miRNA levels change in response to changes in the level of ceRNA j ($g_{\mu j}$). For $g_{i\mu}$, after some algebra one finds that

$$g_{i\mu}(\omega) = - \frac{[m_i]}{[\mu] + \mu_{0,i} z_i(\omega)} , \quad (21)$$

$$z_i(\omega) = \frac{(1 + i\omega\tau_{1,i})(1 + i\omega\tau_{2,i})}{1 + i\omega\tau_{3,i}} , \quad (22)$$

where the timescales $\tau_{1,i}$, $\tau_{2,i}$ and $\tau_{3,i}$ are defined as

$$\tau_{1,i} = 1/d_i , \quad \tau_{2,i} = 1/(k_i^- + \kappa_i + \sigma_i) , \quad \tau_{3,i} = 1/(\kappa_i + \sigma_i) . \quad (23)$$

One sees that

$$\lim_{\omega \rightarrow 0} \frac{g_{i\mu}(\omega)}{g_{i\mu}(0)} = 1 , \quad \lim_{\omega \rightarrow \infty} \frac{g_{i\mu}(\omega)}{g_{i\mu}(0)} = 0 . \quad (24)$$

In other terms, the ratio $g_{i\mu}(\omega)/g_{i\mu}(0) \equiv J_i(\omega)$ acts effectively as a low-pass filter that transmits perturbations whose frequencies are smaller than a certain cut-off frequency (e.g. sufficiently sustained transcriptional upregulations), while it does not transmit higher-frequency perturbations (e.g. short transcriptional bursts). The cut-off frequency is clearly related to the timescales $\tau_{1,i}$, $\tau_{2,i}$ and $\tau_{3,i}$, which means that it can be adjusted by modulating the kinetic parameters that control them.

On the other hand, the form of $g_{\mu j}$ depends on whether $\sigma_j > 0$ or not. In the former case, one obtains

$$g_{\mu j}(\omega) = \frac{\frac{\sigma_j}{\kappa_j + \sigma_j} \frac{1 + i\omega\tau_{4,j}}{1 + i\omega\tau_{3,j}} \left(1 + \frac{\mu_{0,j} z_j(\omega)}{[\mu]}\right)^{-1}}{\delta + i\omega + \sum_i [m_i] k_i^+ \left(1 + \frac{[\mu]}{\mu_{0,i} z_i(\omega)}\right)^{-1}} , \quad (25)$$

where

$$\tau_{4,j} = 1/\sigma_j . \quad (26)$$

As for $g_{i\mu}$, we have

$$\lim_{\omega \rightarrow 0} \frac{g_{\mu j}(\omega)}{g_{\mu j}(0)} = 1 , \quad \lim_{\omega \rightarrow \infty} \frac{g_{\mu j}(\omega)}{g_{\mu j}(0)} = 0 , \quad (27)$$

suggesting that $g_{\mu j}(\omega)/g_{\mu j}(0) \equiv L_j(\omega)$ also behaves like a low-pass filter. This means that, for $\sigma_j > 0$,

$$\widehat{\chi}_{ij}(\omega) \equiv g_{i\mu}(\omega)g_{\mu j}(\omega) = J_i(\omega)g_{i\mu}(0)L_j(\omega)g_{\mu j}(0) \equiv J_i(\omega)L_j(\omega)\widehat{\chi}_{ij}(0) \quad , \quad (28)$$

where both J_i and L_j are low-pass filters (and we remind that $\widehat{\chi}_{ij}(0)$ corresponds to the steady-state susceptibility). In summary: if $\sigma_j > 0$, a dynamical ceRNA crosstalk is expected to be observed for sufficiently long timescales, whereas high-frequency perturbations will instead be attenuated. The magnitude of the dynamical ceRNA effect relative to the steady-state case is given by the product $J_i(\omega)L_j(\omega)$. It can be shown that this product can exceed 1 (i.e. the dynamical response is going to be stronger than the stationary response) when k_i^- is sufficiently large [30]. One can also check that $J_iL_j \rightarrow 1$ (i.e. that $\widehat{\chi}_{ij}(\omega) \rightarrow \widehat{\chi}_{ij}(0)$) as $\omega \rightarrow 0$, which recovers the steady state crosstalk picture.

If $\sigma_j = 0$, instead, one finds

$$g_{\mu j}(\omega) = \frac{\frac{i\omega\tau_{5,j}}{1+i\omega\tau_{5,j}} \left(1 + \frac{\mu_{0,j}z_j(\omega)}{[\mu]}\right)^{-1}}{\delta + i\omega + \sum_i [m_i]k_i^+ \left(1 + \frac{[\mu]}{\mu_{0,i}z_i(\omega)}\right)^{-1}} \quad , \quad (29)$$

where

$$\tau_{5,j} = 1/\kappa_j \quad . \quad (30)$$

This case is radically different from the previous one: because the quantity $\frac{i\omega\tau_{5,j}}{1+i\omega\tau_{5,j}}$ describes a high-pass filter (i.e., contrary to J_i and L_j , it vanishes for $\omega \rightarrow 0$ and tends to 1 for $\omega \rightarrow \infty$), Eq. (29) is the product of a high-pass filter and a low-pass one. This yields a filter $M_j(\omega) \equiv g_{\mu j}(\omega)/g_{\mu j}(0)$ that transmits at intermediate frequencies as opposed to the low frequencies that matter when $\sigma_j > 0$. This clarifies that, when $\sigma_j = 0$, crosstalk is necessarily absent at stationarity but it can be non-zero at some intermediate timescales. The analog of (28) for this case reads

$$\widehat{\chi}_{ij}(\omega) = J_i(\omega)M_j(\omega)\widehat{\chi}_{ij}^c(0) \quad , \quad (31)$$

where the ‘reference’ asymptotic susceptibility is now given by

$$\widehat{\chi}_{ij}^c(0) = \lim_{\sigma_j \rightarrow 0} \frac{\kappa_i + \sigma_i}{\sigma_i} \widehat{\chi}_{ij}(0) \quad (32)$$

(contrary to the steady-state susceptibility $\widehat{\chi}_{ij}(0)$ which is bound to vanish when $\sigma_i = 0$, the quantity $\frac{\kappa_i + \sigma_i}{\sigma_i} \widehat{\chi}_{ij}(0)$ can be different from zero when $\sigma_j \rightarrow 0$). To sum up: if $\sigma_j = 0$, a dynamical ceRNA crosstalk is expected to occur on intermediate timescales, whereas sufficiently high- and low-frequency perturbations are attenuated. Its magnitude relative to the relevant steady-state susceptibility is equal to the product $J_i(\omega)M_j(\omega)$, which can be shown to be close to 1 (i.e. the dynamical response is comparable to the stationary one) when k_i^- is sufficiently small [30].

As we anticipated, the exact ranges of frequencies (i.e. timescales) where transient responses are expected to be observed depend on the kinetic parameters via a few characteristic timescales of the dynamics, namely $\tau_{1,i}$, $\tau_{2,i}$ and $\tau_{3,i}$ (see (23)), $\tau_{4,i}$

(see (26)) and $\tau_{s,i}$ (see (30)). In addition, perhaps unexpectedly, dynamical crosstalk can be more intense than stationary crosstalk, i.e. $\hat{\chi}_{ij}(\omega)/\hat{\chi}_{ij}(0)$ can exceed 1. A careful analysis of the filters introduced above [30] moreover shows that, in certain regimes of kinetic parameters, these effects are not limited to occur between the same ceRNA categories that enjoy a non-zero stationary crosstalk: dynamical susceptibilities can be non-zero even for pairs of free or bound ceRNAs. All in all, the fact that dynamical crosstalk can occur in cases where stationary crosstalk can't clearly says that miRNA-mediated RNA regulation is more pervasive and common than steady state theories would suggest.

3.3 Large perturbations and extended ceRNA effect

The above results have been obtained under the assumptions that perturbations are 'small', i.e. such that deviations from the steady state are linear in the perturbation size. The (possibly more physiologically relevant) case of 'large perturbations' generating non-linear responses unfortunately can only be analysed via simulations. The computational study performed in [30] has shown that, rather remarkably, non-linear effects appear to become more significant when the fold-size of the perturbation exceeds a threshold that increases as the miRNA transcription rate β increases. In such cases, ceRNA crosstalk occurs systematically, i.e. independently of whether ceRNAs are repressed, susceptible or free with respect to miRNAs. In other terms, ceRNA crosstalk is a robust, parameter-independent feature of miRNA-mediated regulatory networks when perturbations are sufficiently large. In particular, a large enough change in a transcription rate can represent a very effective (and low-cost) way for cells to rapidly modify the level of a ceRNA.

A concrete example of the latter point arises in the context of early myogenesis in humans [52]. In short, a major regulatory checkpoint in this process is controlled by a miRNA species (miR-133) that binds both a transcription factor of muscle-specific genes (Mastermind-like protein 1, or MAML1) and a muscle-specific long non-coding RNA (linc-MD1). Linc-MD1 and MAML1 are therefore ceRNAs: as the former sponges miR-133, MAML1 is free to activate the expression of genes essential for muscle-cell differentiation. Both miR-133 and linc-MD1 are however synthesised from a single precursor species (pre-miR-133b). Whether pre-miR-133b is processed into miR-133 or linc-MD1 depends on another RNA, namely the messenger RNA of the RNA-binding human antigen R protein (HuR). Specifically, the probability that pre-miR-133b generates linc-MD1 increases as the level of free HuR increases. HuR however is also a target of miR-133, i.e. it is a ceRNA of both linc-MD1 and MAML1. As a consequence, high levels of HuR strongly favour high levels of linc-MD1 and therefore enhance the expression of the muscle-specific genes controlled by MAML1. Such a situation amounts to a self-reinforcing regulatory feed-forward loop that expresses the muscle-specific genes regulated by MAML1. This loop however has to be silenced in order for differentiation to transition into the subsequent stage. Cells achieve this through a large, rapid, endogenous increase

in miR-133 levels driven by the activation of a second channel of miRNA transcription, independent of pre-miR-133b, that induces the fast down-regulation of MAML1, linc-MD1 and HuR. Such a mechanism has been studied experimentally in [52] and mathematically in [32]. In mathematical terms, the activation of the second miRNA biosynthesis channel corresponds to a step-like increase in miR-133 transcription rate occurring at a certain time t^* . This can be described by a time-dependent miRNA transcription rate of the form

$$\beta(t) = \beta_0 [1 + \alpha \theta(t - t^*)] = \begin{cases} \beta_0 & \text{for } t < t^* \\ \beta_0(1 + \alpha) & \text{for } t > t^* \end{cases}, \quad (33)$$

where β_0 is miR-133's baseline transcription rate (before the perturbation), α represents the fold-change in β that occurs at t^* , and $\theta(x) = 1$ if $x > 0$ and 0 otherwise. To quantify the response to this perturbation, one can analyze (numerically) the quantity

$$\mathcal{J}(\alpha) = \int_0^\infty [m(t + t^*) - m(t^*)] dt, \quad (34)$$

where $m(t)$ is the level of MAML1 at time t . As argued in [30], $\mathcal{J}$ (called the 'integrated response') is a proxy for the change in the total amount of proteins synthesized from m . (Note that, as in this case m gets downregulated after time t^* , $\mathcal{J}$ is negative.) The analysis carried out in [32] has shown that $\mathcal{J}$ is approximately linear in α when the complex formation kinetics occurs with slow rates. On the other hand, when the kinetics gets faster the linear response regime breaks down and values of α as low as 20% suffice to generate large (non-linear) changes in m . It is possible to extend the theory developed in [30] to calculate the characteristic timescale τ over which the system relaxes to a new steady state following a large perturbation. It turns out that $\tau \propto \alpha/\beta$. This implies that, if kinetic parameters are such that a small α ensures the desired change, such a change can occur over relatively short timescales, as would likely be required in differentiation processes.

In the following section we focus on the ceRNA response to a specific type of perturbation, namely one in which miRNA transcription is activated in small periodic bursts.

4 ceRNA crosstalk under periodic miRNA transcriptional pulses in a purely catalytic miRNA-ceRNA network

Periodic signals have emerged as an important mode of regulation in the post-transcriptional context. For instance, some regulatory circuits where miRNAs and transcription factors interact have been found to function in pulses [48, 49]. In some specific pathways, miRNA expression itself has been observed to be periodic in time [48, 51]. A functional role of periodic miRNA-mediated regulation has recently been proposed in [29], where it was shown by ODE modelling that a period-

ically expressed miRNA can lead to a stronger target RNA repression than constant miRNA expression. Moreover, the authors showed that periodic miRNA expression can generate a frequency preference behaviour, meaning that specific input frequencies lead to maximal target RNA repression. The same study investigated the interplay between periodic miRNA expression and competition between RNA targets. In particular, we shall now see that competition between ceRNAs for a periodically expressed miRNA introduces some interesting frequency-dependent regulatory effects, that turn out to be coherent with the theoretical considerations outlined above.

Figure 4 shows the one-miRNA two-ceRNAs model used in [29] to analyze the effects of periodic miRNA synthesis on the fold repression of the two competing targets.

Similarly to the model represented in Figure 2, ceRNAs and miRNAs are transcribed and degraded; in addition, two species of miRNA-ceRNA complexes can be formed; and each molecular species can be degraded both when unbound and when in complex. However, at odds with the model of Figure 2, the model in Figure 4 assumes that ceRNAs and miRNA are degraded independently when in complex, as suggested by previous data [4, 20, 37]: either the RNA is degraded and the miRNA is recycled into the system, or vice versa. With this modification, the kinetic equations take the form

$$\begin{aligned}\frac{dm_i}{dt} &= b_i - d_i m_i - k_i^+ m_i \mu + (k_i^- + \lambda_i) c_i , \\ \frac{d\mu}{dt} &= \beta - \delta \mu - \sum_i k_i^+ m_i \mu + \sum_i (k_i^- + \kappa_i) c_i , \\ \frac{dc_i}{dt} &= k_i^+ m_i \mu - (k_i^- + \lambda_i + \kappa_i) c_i ,\end{aligned}\tag{35}$$

where the index $i = 1, 2$ identifies either of the two ceRNA species and the additional parameter λ_i represents the rate of miRNA degradation in complex with ceRNA i (implying recycling of ceRNA i). Note that no explicit stoichiometric degradation rate appears for complexes. In order to feed the model with a periodic miRNA synthesis signal, one can assume that the miRNA synthesis rate constant β is described by a square wave, i.e.

$$\beta_{pulse}(t) = \begin{cases} \beta_0 & \text{if } nT < t \leq (n+D)T, \quad n = 0, 1, 2, \dots \\ 0 & \text{if } (n+D)T < t \leq (n+1)T, \quad n = 0, 1, 2, \dots \end{cases}\tag{36}$$

where β_0 is the amplitude, T is the period, and $D \leq 1$ is the duty cycle, i.e. the ratio between the pulse width and its period. Upon some inspection, one notices that the model can be re-written in terms of the dimensionless variables

$$\hat{t} = d_1 t, \quad \hat{m}_i = \frac{m_i}{b_1/d_1}, \quad \hat{\mu} = \frac{\mu}{b_1/d_1}, \quad \hat{c}_i = \frac{c_i}{b_1/d_1} .\tag{37}$$

Here, $\hat{m}_i$ represents the dimensionless concentration of the unbound i -th ceRNA species, $\hat{\mu}$ is the dimensionless concentration of unbound miRNA, and $\hat{c}_i$ represent

the dimensionless concentration of the i -th complex. As a result of this scaling, one time unit ($\hat{t} = 1$) corresponds to $d_1 t = \frac{\ln(2)}{h} t = 1$ or $t = h / \ln(2)$, where h is the half-life of ceRNA 1. In other terms, one unit of re-scaled time corresponds to approximately to 1.44 ceRNA1 half-lives. We thus obtain the re-scaled system

$$\begin{aligned} \frac{d\hat{m}_1}{d\hat{t}} &= 1 - \zeta_1^+ \hat{m}_1 \hat{\mu} + \zeta_1^- \hat{c}_1 - \hat{m}_1 + \gamma \varphi_1 \hat{c}_1 , \\ \frac{d\hat{m}_2}{d\hat{t}} &= \rho \hat{m}_2 - \zeta_2^+ \hat{m}_2 \hat{\mu} + \zeta_2^- \hat{c}_2 - \varepsilon \hat{m}_2 + \gamma \varphi_2 \hat{c}_2 , \\ \frac{d\hat{\mu}}{d\hat{t}} &= \eta - \zeta_1^+ \hat{m}_1 \hat{\mu} + \zeta_1^- \hat{c}_1 - \gamma \hat{\mu} + \alpha_1 \hat{c}_1 - \zeta_2^+ \hat{m}_2 \hat{\mu} + \zeta_2^- \hat{c}_2 + \alpha_2 \hat{c}_2 , \\ \frac{d\hat{c}_1}{d\hat{t}} &= \zeta_1^+ \hat{m}_1 \hat{\mu} - \zeta_1^- \hat{c}_1 - \gamma \varphi_1 \hat{c}_1 - \alpha_1 \hat{c}_1 , \\ \frac{d\hat{c}_2}{d\hat{t}} &= \zeta_2^+ \hat{m}_2 \hat{\mu} - \zeta_2^- \hat{c}_2 - \gamma \varphi_2 \hat{c}_2 - \alpha_2 \hat{c}_2 , \end{aligned} \quad (38)$$

with new kinetic parameters that are now ratios of the original rate constants, namely

$$\begin{aligned} \eta &= \beta / b_1 , \quad \gamma = \delta / d_1 , \quad \rho = \frac{b_2}{b_1} , \quad \varepsilon = \frac{d_2}{d_1} , \\ \zeta_i^+ &= \frac{k_i^+ b_1}{d_1^2} , \quad \zeta_i^- = \frac{k_i^-}{d_1} , \quad \alpha_i = \frac{\kappa_i}{d_1} , \quad \varphi_i = \frac{\lambda_i}{\delta} . \end{aligned} \quad (39)$$

The above rates have the following physical meanings: η represents the synthesis rate constant of miRNA relative to that of ceRNA 1; γ is the degradation rate of unbound miRNA relative to that of unbound ceRNA 1; ρ and ε describe respectively the synthesis and the degradation rates of ceRNA 2 m_2 relative to ceRNA 1 m_1 ; ζ_i^+ and ζ_i^- represent respectively the scaled binding and unbinding rates of the i -th ceRNA to the miRNA; α_i describes the degradation rate of ceRNA i in complex relative to that of its unbound form; and φ_i is the degradation rate of the miRNA in complex with the i -th ceRNA relative to its unbound form. Correspondingly, the dimensionless periodic miRNA synthesis signal is given by the square wave

$$\eta_{pulse}(t) = \begin{cases} \eta_0 = \beta_0 / b_1 & \text{if } nT < t \leq (n+D)T, \quad n = 0, 1, 2, \dots \\ 0 & \text{if } (n+D)T < t \leq (n+1)T, \quad n = 0, 1, 2, \dots \end{cases} \quad (40)$$

where η_0 is now the dimensionless amplitude. Looking back at the dimensional model, the amounts of ceRNA1 and ceRNA2 synthesized in one period of the square wave input would correspond respectively to

$$\int_0^T b_1 dt = b_1 T \quad \text{and} \quad \int_0^T b_2 dt = b_2 T , \quad (41)$$

whereas the miRNA dose synthesized in one period would be

$$\int_0^T \beta_{pulse} dt = \beta_0 T D . \quad (42)$$

In terms of the dimensionless parameters, the amounts of miRNA synthesized in one period relative to ceRNA1 and, respectively, ceRNA2 now read

$$\frac{\beta_0 T D}{b_1 T} = \eta_0 D \quad \text{and} \quad \frac{\beta_0 T D}{b_2 T} = \frac{\eta_0}{\rho} D \quad (43)$$

respectively. The quantities are independent from the square wave's frequency. For a fixed parameter set, it is thus sufficient to keep the square wave's duty cycle D constant, which we fix to 0.5, to maintain a constant the amount of repressor synthesized relative to each ceRNA in a complete pulse. In this way, conservation of the miRNA-ceRNA relative dose produced in a given time interval Δt is ensured for any frequency that accomodates an integer number of pulses in such an interval, i.e. frequencies that satisfy $\text{mod}(\Delta t/T)$. It is thus possible to feed the model either with an appropriate periodic miRNA synthesis rate η_{pulse} or with a constant synthesis rate with identical relative miRNA-ceRNA dose, $\eta_{const} = \eta_0 D$, as shown in Fig. 5.

As a result of the interplay between the system's timescale and the input's timescale, different ceRNA dynamics can be found, from periodic to rising dynamics. With the aim of comparing the average level of repression of the two ceRNAs in the same timespan, one can analyze the time τ needed by one ceRNA to reach steady state when constantly stimulated (i.e. its equilibration time), and then assess how the two ceRNAs respond to different frequencies of miRNA synthesis in such an interval. Note that, to satisfy the dose conservation criterion, the frequency range is limited by the lower bound $1/\tau$, whereas one can choose an arbitrary upper bound of $10^{-2}/\tau$.

In this way it is possible to address whether, in the timespan over which one target equilibrates, one can obtain, for a given frequency, either a concomitant response of the two ceRNAs or an exclusive response by one ceRNA only, or no response by neither of the two. The average fold repression of ceRNA i in the timespan $[0, \tau]$ can be evaluated for constant and periodic input synthesis signals by computing respectively

$$FR_i^{\text{const}} = \frac{\langle \hat{m}_i^c(t) \rangle_{[0, \tau]}}{\langle \hat{m}_i^{\text{const}}(t) \rangle_{[0, \tau]}} \quad (44)$$

and

$$FR_i^{\text{pulse}} = \frac{\langle \hat{m}_i^c(t) \rangle_{[0, \tau]}}{\langle \hat{m}_i^{\text{pulse}}(t) \rangle_{[0, \tau]}} \quad (45)$$

where $\hat{m}_i^c(t)$ represents ceRNA i 's constitutive time evolution (i.e. its temporal expression achieved in the absence of miRNA), and $\hat{m}_i^{\text{const}}(t)$ and $\hat{m}_i^{\text{pulse}}(t)$ represent ceRNA i 's time evolution in presence of miRNA produced by the constant input synthesis η_{const} or by the periodic input synthesis η_{pulse} , respectively. Note that the ratio makes these quantities independent of the scaling used to make the model dimensionless. The emergent scenario can be summarized as follows.

4.1 Frequency preference and advantage of periodic over constant input

By exploring the parameter space of the model, it is possible to show that –when stimulated with periodic miRNA synthesis– each of the two ceRNAs can either under- or outperform its fold repression given by constant synthesis [29]. Moreover, depending on the specific ceRNA kinetics, a given miRNA input frequency can generate higher or lower fold repression, exhibiting a band-pass filtering-like behaviour. This is in full agreement with the general theoretical considerations outlined in the previous section.

It is of particular interest to address cases where both competitors display a frequency preference, i.e. where each ceRNA is maximally repressed for a different frequency of the input (Figure 6). This scenario describes a situation where optimal regulation for each ceRNA is achieved with a distinct temporal pattern of miRNA synthesis, and thus might represent a way of selective target regulation in a ceRNA network. The next section summarizes the kinetic conditions where this kind of scenario is achieved.

4.2 Selective ceRNA regulation by desynchronization in frequency

To investigate the role of kinetics in the emergence of selective scenarios –i.e. where the two ceRNAs are optimally repressed by distinct frequencies (Figure 7)– one can sample the model’s parameter space and correlate parameter values with the occurrence of band-pass filtering behaviours [29]. With this approach, it is possible to show that the most relevant timescales for the occurrence of frequency-dependent selective repression are those of miRNA-ceRNA binding and those of relative degradation of the two competitors. In particular, the most relevant parameter resulted to be the relative dissociation constant, which in the dimensionless model is given by $\frac{Z_2}{Z_1} = \frac{\zeta_2^+}{\zeta_2^-} \cdot \frac{\zeta_1^-}{\zeta_1^+}$. It appears indeed that when the ceRNA-miRNA binding affinities differ greatly, the two ceRNAs are more likely to present optimal repression at distinct frequencies.

Similarly, the ratio α_2/α_1 (i.e. the ratio between ceRNA2 and ceRNA1 degradation rates in complex relative to their unbound form) and the parameter ε (i.e. the relative decay rate of the two ceRNAs) appear correlated with the selective frequency-dependent behavior. This means that the separation of respective degradation timescales of the two ceRNAs is also determinant for the emergence of exclusive frequency-driven repression.

It is thus speculated that periodic miRNA expression of suitable frequency might allow to exclusively regulate single ceRNAs out of a pool of multiple targets - if their reciprocal kinetics are properly balanced.

5 Outlook

Compared to the stationary crosstalk scenario, the dynamical picture is radically different in two aspects: first, transient crosstalk can occur between ceRNAs independently of whether they are asymptotically repressed, susceptible or unrepressed with respect to their miRNA regulators; second, if kinetic parameters allow for it, transient crosstalk can be more intense than its stationary counterpart. Overall, dynamical ceRNA crosstalk should therefore be expected to be an inherent, unavoidable and pervasive part of post-transcriptional regulation.

On the other hand, ceRNA crosstalk appears to lose two of its key steady-states features during transients. First is selectivity: stationary crosstalk is highly selective [31, 61], that is, by tuning kinetic parameters and miRNA levels, miRNA-ceRNA networks can channel signals to specific nodes while avoiding general spillover of perturbations that could be damaging. This feature appears to be lost dynamically as ceRNA crosstalk patterns are more weakly dependent on kinetic parameters. While one may argue that some degree of selectivity is preserved dynamically, it is important to note that a different kind of selectivity is active during transient, namely selectivity to frequency. The possibility to respond to certain time-dependent signals and not others is a major advantage when regulatory processes are strictly time-resolved (as e.g. in development).

The second feature that appears to be lost dynamically is efficient noise processing. One of the key findings of steady-state theories has involved the realization that miRNA-ceRNA systems are capable of efficiently processing the natural variability of signals, both intrinsic and extrinsic [16, 39, 59]. It is not clear how much of this capability is preserved away from the steady state. Modeling work in this direction would be highly desirable. However, one may expect dynamical noise-processing to be maximally efficient when the characteristic timescales of the dynamics and of the noise are not too different. Indeed, if noise is too fast compared to molecular levels, it will effectively be washed out even during transients; while if noise has long enough characteristic timescales, they will conflict with the equilibration timescale of molecular levels (i.e. the dynamics will stabilize fast than the noise affects it). As a consequence, optimal noise processing requires some matching of timescales. Because it is reasonable to think that molecular levels will react to small noise in the same way as they react to small perturbations, it is reasonable to think that noise frequency is going to be a key modulator of the dynamical noise-processing capacity of miRNA-ceRNA networks.

The main challenge towards a deeper theoretical understanding of the dynamical features of ceRNA crosstalk lies, in our view, in exporting the theory developed for small network to large-scale (genome-level) regulatory network. Lack of knowledge about interactions and constants makes this a particularly daunting task. Statistical methods such as those employed e.g. in [61] may however help elucidate at least the conditions under which robust global expression profiles and miR-programs can emerge dynamically.

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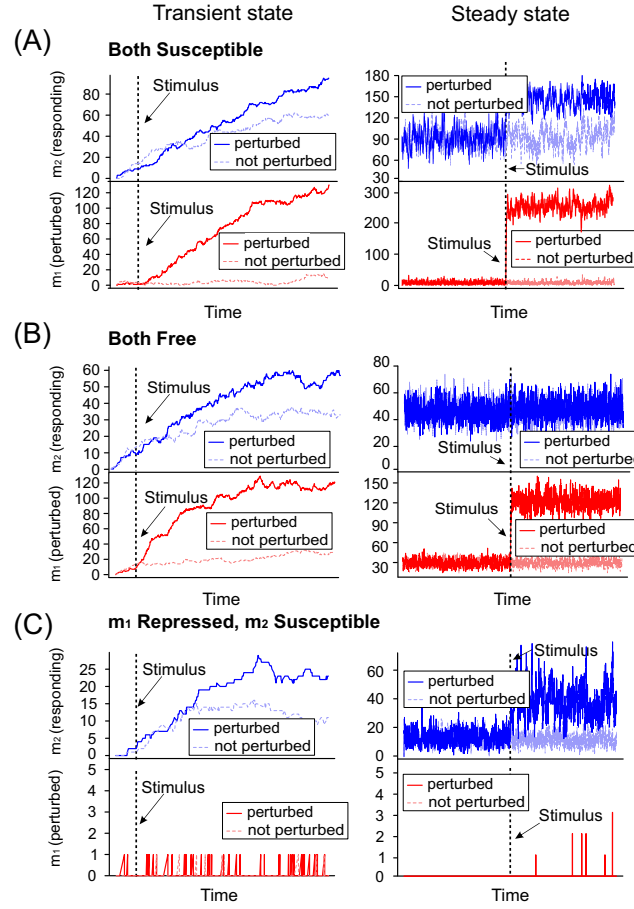


Fig. 1 Crosstalk scenario for a model with two ceRNAs interacting through a shared miRNA upon perturbation of one ceRNA's transcription rate. Panels on the left show the behaviour within the transient while panels on the right show the behaviour at the steady state. Dashed black line in each plot indicates the time point at which the transcription rate of ceRNA 1 m_1 is increased. (A) Both ceRNAs are in their repressed regime. Upon punctual increasing of m_1 transcription rate, m_2 amount increases, both within the transient (left panel) and at the steady state (right panel). (B) Both ceRNAs are in their free regime. Upon punctual increasing of m_1 transcription rate, m_2 responds only upon m_1 perturbation within the transient (left panel), while it is insensitive to perturbation at the steady state (right panel). (C) ceRNA 1 m_1 is in its repressed regime and ceRNA 2 m_2 is in its susceptible regime. Upon punctual increasing of m_1 transcription rate, m_2 responds both within the transient (left panel) and at the steady state (right panel). In all the panels corresponding to the transient state, the signal-to-noise ratio is > 1 .

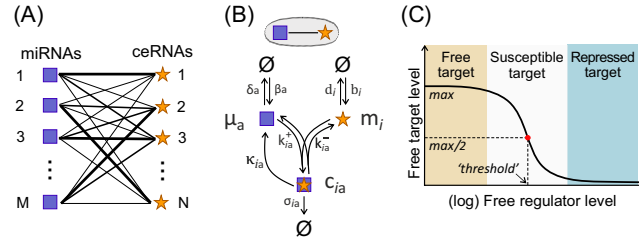


Fig. 2 (A) Generic network of M miRNAs interacting with N ceRNAs. The thickness of the links represents the different interaction strengths between the molecular species. (B) Focusing on the interaction between a miRNA μ_a and a ceRNA m_i , the panel shows the kinetic rates associated to the reactions considered. (C) Sketch of the three stoichiometric regimes that it is possible to identify for the free amount of target (ceRNA) as a function of the free amount of the regulator (miRNA).

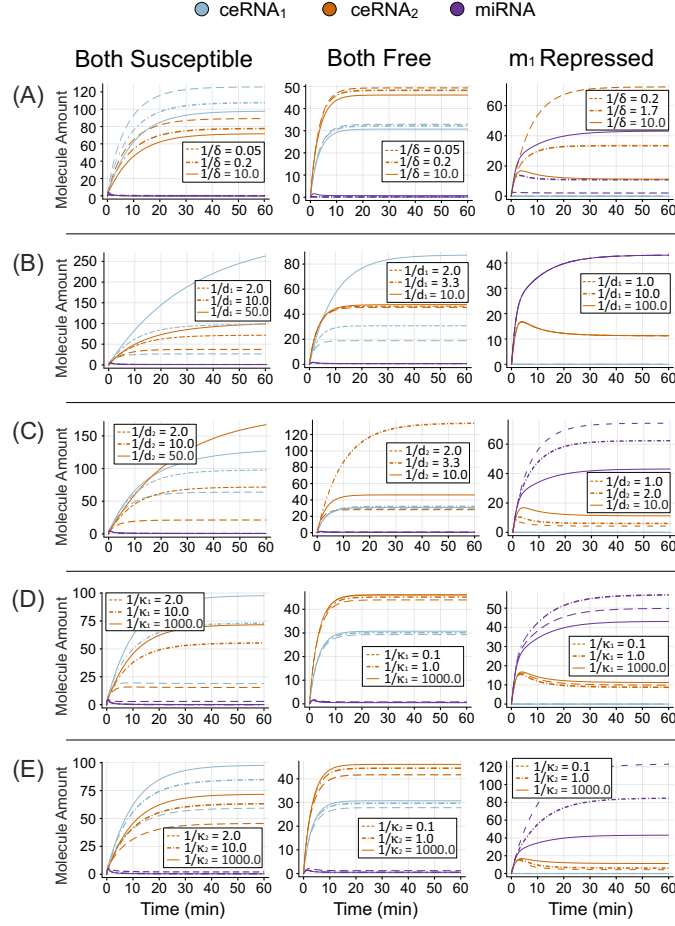


Fig. 3 Dynamical behaviour over time of a model considering two ceRNAs competing for one miRNA. CeRNA1 in blue, ceRNA2 in orange, and miRNA in purple as function of time for different values of the mean miRNA lifetime $1/\delta$ in panel (A), the mean ceRNA1 or ceRNA2 lifetimes, $1/d_1$ or $1/d_2$ in panels (B) and (C), respectively, the mean time needed for the two complexes catalytic degradation, $1/\kappa_1$ or $1/\kappa_2$ in panels (D) and (E), respectively. The plots are organized depending on the different regimes reached at the steady state and defined in (3). From left to right: both ceRNAs are in their susceptible regime (left), both ceRNAs are in their unrepressed or free regime (center), ceRNA1 is in its repressed regime while ceRNA2 is susceptible (right). Lifetimes in all the legends are in minutes.

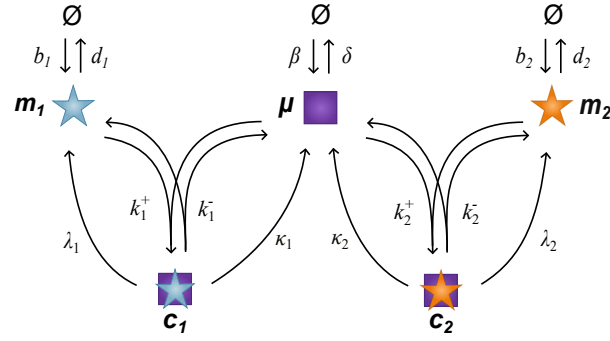


Fig. 4 Network of one miRNA interacting with two ceRNAs. m_1 and m_2 represent the two ceRNA species, μ represents the miRNA, and c_1 and c_2 represent molecular complexes formed respectively by ceRNA1 and ceRNA2 with the miRNA. b_1 and b_2 describe respectively the synthesis rates of ceRNA1 and ceRNA2, while d_1 and d_2 are their respective degradation rates. β and δ represent the miRNA synthesis and degradation rates. k_1^+ and k_2^+ represent the two ceRNA-to-miRNA binding rates, whereas k_1^- and k_2^- describe their respective unbinding rates. κ_1 and κ_2 are the respective degradation rates of ceRNA1 and ceRNA2 in complex with the miRNA, λ_1 and λ_2 are miRNA degradation rates in complex respectively with ceRNA1 and ceRNA2.

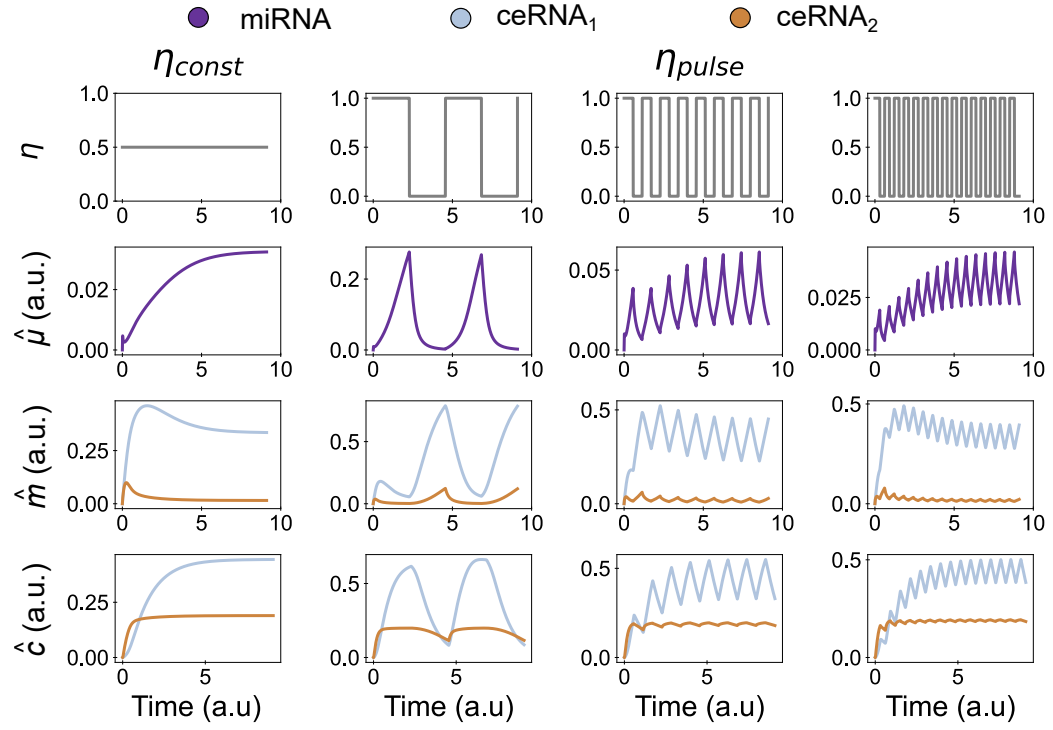


Fig. 5 Dynamical behaviour over time of two ceRNAs competing for a constantly or periodically synthesized miRNA. The first column of plots shows the constant input miRNA synthesis rate (top plot), and the resulting timecourses of miRNA (middle top plot, violet curve), ceRNAs (middle bottom plot, light blue and dark orange curves) and the two complexes (bottom plot, light blue and dark orange curves). The remaining three columns of plots represent periodic input miRNA synthesis rates with three distinct frequencies (top plots), and the resulting timecourses of miRNA (middle top plots, violet curves), ceRNAs (middle bottom plots, light blue and dark orange curves) and the two complexes (bottom plots, light blue and dark orange curves).

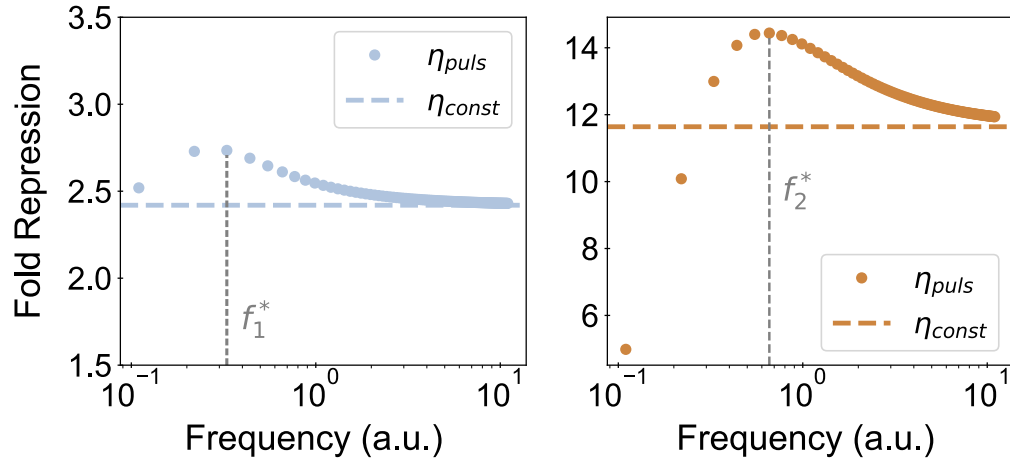


Fig. 6 Fold repression for two ceRNA species computed over the interval $[0, \tau]$ as a function of the dimensionless miRNA synthesis frequency. The left plot represents $FR_1^{pulse}(f)$ (light blue dots) and FR_1^{const} (light blue dashed line). The vertical dashed line indicates the preferred frequency f_1^* . The right plot represents $FR_2^{pulse}(f)$ (dark orange dots) and FR_2^{const} (dark orange dashed line). The vertical dashed line indicates the preferred frequency f_2^* .

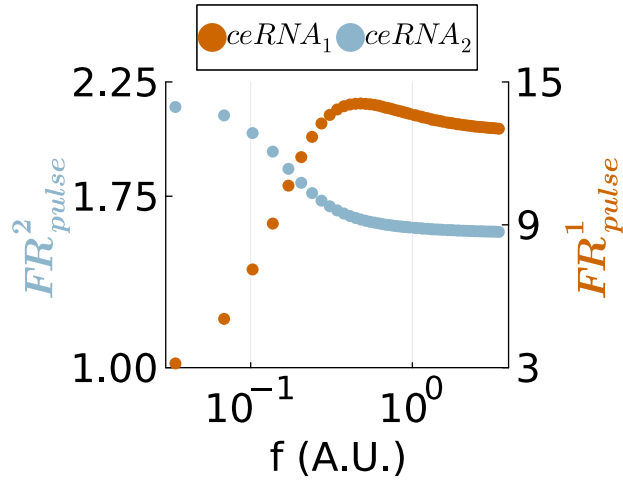


Fig. 7 Fold repression as a function of miRNA synthesis frequency for two ceRNAs in a selective regulation case - where the two species are optimally repressed at distinct frequencies. The two curves represent FR_1^{pulse} (dark orange) and FR_2^{pulse} (light blue) as a function of the dimensionless frequency f .

Table 1 Basal parameter values for Figure 1, panels (A-C)

Parameters	Both susc.	Both free	m_1 repr. m_2 susc.	Units
b_1	10.0	10.0	2.0	molecule min^{-1}
b_2	15.0	15.0	10.0	molecule min^{-1}
β	15.0	2.0	15.0	molecule min^{-1}
k_1^+	1.0	10^{-3}	10^5	molecule $^{-1}$ min^{-1}
k_1^-	10^{-3}	10^{-3}	10^{-3}	min^{-1}
k_2^+	10^{-3}	10^{-3}	10^{-4}	molecule $^{-1}$ min^{-1}
k_2^-	10^{-3}	10^{-3}	10^{-3}	min^{-1}
κ_1	10^{-3}	10^{-3}	0.1	min^{-1}
κ_2	10^{-3}	10^{-3}	10^{-3}	min^{-1}
σ_1	1.0	1.0	1.0	min^{-1}
σ_2	1.0	1.0	1.0	min^{-1}
d_1	0.1	0.3	0.1	min^{-1}
d_2	0.1	0.3	0.1	min^{-1}
δ	0.1	0.1	0.1	min^{-1}

Table 2 Basal parameter values for Figure 3, panels (A-E)

Parameters	Both susc.	Both free	m_1 repr. m_2 susc.	Units
b_1	24.0	10.0	2.0	molecule min^{-1}
b_2	15.0	15.0	10.0	molecule min^{-1}
β	22.0	2.0	15.0	molecule min^{-1}
k_1^+	0.4	10^{-3}	10^5	molecule $^{-1}$ min^{-1}
k_1^-	10^{-3}	10^{-3}	10^{-3}	min^{-1}
k_2^+	0.3	10^{-3}	10^{-4}	molecule $^{-1}$ min^{-1}
k_2^-	10^{-3}	10^{-3}	10^{-3}	min^{-1}
κ_1	10^{-3}	10^{-3}	0.1	min^{-1}
κ_2	10^{-3}	10^{-3}	10^{-3}	min^{-1}
σ_1	0.3	1.0	1.0	min^{-1}
σ_2	0.3	1.0	1.0	min^{-1}
d_1	0.1	0.3	0.1	min^{-1}
d_2	0.1	0.3	0.1	min^{-1}
δ	0.1	0.1	0.1	min^{-1}

Table 3 Parameter values for Figure 5, Figure 6, and Figure 7

Dimensionless parameters	Values Figs. 5-6	Values Fig.7
η	1.0	1.0
γ	0.75	0.25
ρ	1.5	1.0
ε	5.0	1.6
ζ_1^+	500.0	190.0
ζ_1^-	5000.0	2.5
ζ_2^+	10.0	6025.0
ζ_2^-	5.0	2.5
α_1	1.5	1.4
α_2	7.5	1.4
φ_1	1.0	1.4
φ_2	1.0	1.4